

UCL Electrical Competence Group

Technical Safety Framework & Decision Matrix

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1 Introduction

This document outlines the safety framework for electrical activities within UCL laboratories. It categorizes work based on risk levels and provides a standardized path for documentation, inspection, and approval.

2 Activity Classification

Activities are divided into two primary categories to streamline safety reviews:

Cat 1: Standard Work. Use of off-the-shelf, CE-marked equipment connected via standard plugs to MCB-protected circuits for their intended purpose.

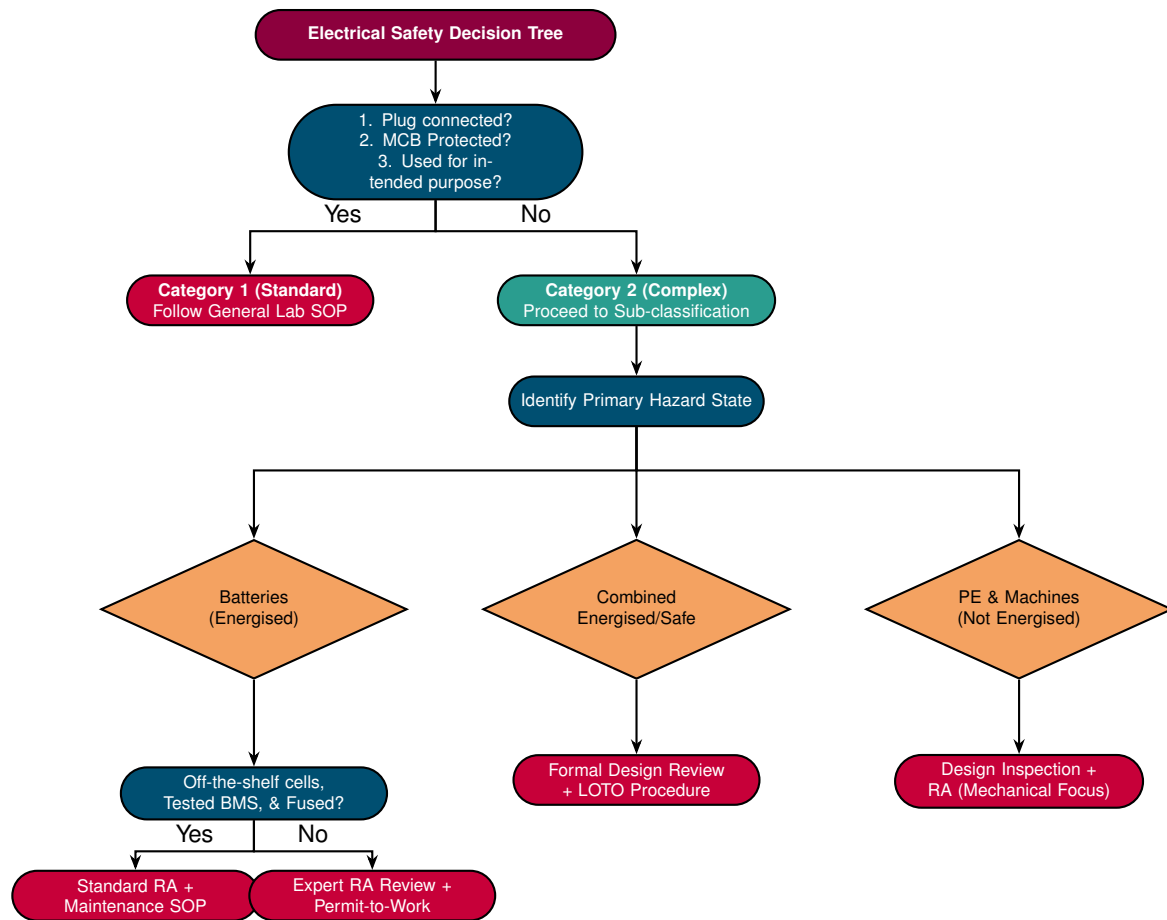
Cat 2: Complex/Experimental Work. Includes bespoke assemblies, battery pack construction, and modified power electronics.

Activities exceeding 42.2V AC or 60V DC are automatically classified as Category 2 and require a Specialist RA Review regardless of power level.

2.1 Category 2 Subcategories

1. **Batteries (Energised):** Ranging from assembly of known-good cells (Type A) to experimental chemistries (Type B).
2. **Power Electronics & Machines (Non-Energised):** Construction, mechanical assembly, and wiring phases.
3. **Simultaneously Energised/Non-Energised:** Complex test-beds where high-voltage and low-voltage control circuits coexist.

3 UCL Activity Decision Tree



A mandatory second person is required for energised work.

4 Maintenance & Documentation

All Category 2 research must include a **Maintenance Plan** within the Risk Assessment (RA).

- **SOPs:** Must detail startup, shutdown, and emergency isolation (LOTO).
- **Regular Checks:** Weekly visual inspection of cable insulation and terminal tightness.
- **Modification Log:** Any change to the circuit requires a secondary sign-off.
- **Decommissioning:** RAs must include a safe disassembly plan to ensure batteries are discharged, and systems are made safe before disposal or storage.

5 Technical Standards for Experimental Setups

To ensure "Protected Upstream" compliance, the following must be met:

- **MCBs:** Must be sized for inrush current. Type B for resistive loads, Type C/D for inductive machines and power electronics.
- **Plugs:** No "daisy-chaining." High-current devices (>10A) must use dedicated wall sockets.
- **Earthing:** All metallic enclosures for custom PE must be bonded to the building Earth.
- **Labelling:** All Category 2 setups must display clear signage indicating the primary hazard (e.g., 'High Voltage') and the names of authorized operators.

6 Review Checklist (RA / Permit to Work)

Item	Requirement
Termination	Crimped or ferruled; no stray strands.
Cable Rating	Current capacity matches max fuse/MCB rating.
Competency	Operator has "Cat 2 Research" training or equivalent.
Protection	Accessible E-Stop and active BMS with mandatory cut-offs for over-temperature, over-voltage
Secondary Risk	Thermal management (cooling) and fire suppression nearby.
Hazardous Env.	Assessed for explosive gases (e.g., battery off-gassing) or wet conditions; ATEX or IP67+ con
Lone Working	Forbidden for energised Cat 2 work; a briefed "Second Person" must be present/available.

7 Emergency Response

In the event of an electrical incident, isolate power at the nearest MCB/Emergency Stop and contact a first-aider immediately.

8 Exemptions & Policy

- **Exemptions:** Low energy systems (<42.2V AC or <60V DC) driven by current-limited sources (e.g., standard bench power supplies) are exempt from the Specialist Review but still require a basic Lab RA.
- **Battery Exclusion:** This exemption does **not** apply to bare battery cells or custom packs. Due to the severe risks of high short-circuit currents and thermal runaway, all energized battery assembly work, regardless of voltage, remains Category 2.
- **CPD Training:** Currently, UCL lacks a specialised course for research into electrical safety. It is strongly proposed that a "Research Electrical Safety" course be developed to ensure competency and safety across complex lab activities.